

Lab sheet 02**Submission Due: 12.06.2024****Title:** Introduction to PHP - II.**Aims:**

- Refresh the knowledge on operators.
- Refresh the knowledge on PHP statements.
- Refresh the knowledge on PHP Loops.
- Introduction to PHP functions and arrays.

Tasks:

- Operators in PHP
- PHP Conditional Statements
- PHP Loops
- PHP Functions
- PHP Arrays

Activities

1. Operators in PHP
 - I. Try each operator and display its outputs.

```
<?php
// Arithmetic Operators
// + , - , * , / , % , ++ , --
$number1 = 45;
$number2 = 67;

$sum = $number1 + $number2;
echo "Sum : " . $sum;
?>
```

```
<?php
// Comparison Operators
// == , != , > , < , >= , <=
$number3 = 45;
$number4 = 67;

$is_equal = $number1 == $number2;
echo "Is Equal ( ".$number3.", ".$number4." ) : " . $is_equal;
?>
```

```
<?php
// Logical Operators
// and , or , && , || , !
```

```
$true_val = true;
>false_val = false;

$and_operator = ($true_val and $false_val);
echo "AND ( " . $true_val . ", " . $false_val . " ) : " .
$and_operator;
?>
```

```
<?php
// Assignment Operators
// = , += , -= , *= , /= , %=

$var1 = 23;
$var2 = 34;

$assignment_opr = $var1 + $var2;
echo "Simple Assignment ( " . $var1 . ", " . $var2 . " ) : " .
$assignment_opr;
?>
```

```
<?php
// String Operators
// . , .=

$string_var1 = "Olivia";
$string_var2 = "Edith";

$string_var1 .= $string_var2;
echo $string_var1;
?>
```

```
<?php
// Conditional Operators
// ?:

$number5 = 3;
$number6 = 4;

$cond_output = $number5 == $number6 ? "Number_5 and Number_6 are
equal." : "Number_5 and Number_6 are not equal.";

echo $cond_output;
?>
```

2. Conditional Statements of PHP

```
<?php
// IF STATEMENT
if (condition) {
    // code to be executed if condition is true;
}

// IF-ELSE STATEMENT
if (condition) {
    // code to be executed if condition is true;
} else {
    // code to be executed if condition is false;
}

// IF-ELSEIF-ELSE STATEMENT
if (condition) {
    // code to be executed if this condition is true;
} elseif (condition) {
    // code to be executed if first condition is false and this
condition is true;
} else {
    // code to be executed if all conditions are false;
}
?>
```

- I. If the room temperature is above 72 degrees Fahrenheit, turn on the air conditioner. If the temperature is below 68 degrees Fahrenheit, turn on the heater. Else, maintain the current temperature.

```
<?php
// SWITCH STATEMENT
switch (expression) {
    case label1:
        //code block
        break;
    case label2:
        //code block;
        break;
    case label3:
        //code block
        break;
    default:
        //code block
}
?>
```

- II. Imagine a vending machine that dispenses different snacks based on user selection. The user presses a button corresponding to a specific snack code:

C001 : for Chips
C002 : for Cookies
C003 : for Candy Bar
D : Invalid selection

3. Loops of PHP

```
<?php
//while loop
while (condition) {
    # code...
}

//do.. while loop
do {
    # code...
} while (condition);
?>
```

- I. Create a script that displays 1-2-3-4-5-6-7-8-9-10 on one line. There will be no hyphen(-) at starting and ending position.

```
<?php
//for loop
for (expression1, expression2, expression3) {
    // code block
}
?>
```

- II. Write a for loop that generates a multiplication table for a number 7. The loop should iterate from 1 to 10 and print each number multiplied by 7.

```
<?php
//foreach loop
foreach ($array_var as $value) {
    # code...
}
?>
```

- III. Imagine you have an array containing items in a shopping cart (e.g., ["apples", "bread", "milk"]). Write a foreach loop that iterates through this array and prints the name of each item in the cart.

4. Functions of PHP

```
<?php

function function_name (arguments) {
    # code...
}

function_name(value);

?>
```

- I. Imagine you're writing a program to calculate the total bill amount for a customer's purchase. Here's how you can achieve that using functions:
- Function to calculate the tax amount(`calculate_tax_amount`): This function takes two parameters: `price` and `tax_rate` . It calculates the tax amount by multiplying the price by the tax rate and returns the result.
 - Function to calculate the total bill: This function takes two parameters: `price` and `tax_rate`. It calls the `calculate_tax_amount` function (defined earlier) to get the tax amount for the given price and tax rate. Then, it adds the tax amount to the original price to get the total bill amount and returns the final amount.
 - Calling the functions and displaying the outputs.

5. Arrays of PHP

```
<?php
// To define - can use array() construct or [] syntax
$empty_array1 = array();
$empty_array2 = [];

$numbers1 = array(1, 2, 3);
$numbers2 = [1, 2, 3];

?>
```

- I. Imagine you're managing a shopping cart system where items are stored in an array. Each item is represented as an associative array with properties like "name", "price", and "quantity". You want to implement functions to achieve the following:
- Calculate total per item: This function should take a single item (associative array) as input and calculate the total cost for that specific item by multiplying the "price" by the "quantity". It should return the calculated total cost.
 - Calculate total bill: This function should iterate through the entire shopping cart array containing all items. Within the loop, it should call the previously defined function (calculate total per item) for each item in the cart. It should accumulate the individual item totals and return the final total bill amount.

Discussion:

- Array Operators
- Associative Arrays
- Variadic functions